A credibility-weighted hybrid soft sensor and uncertainty-consuming robust real-time optimization for silica scaling control in Philippine geothermal fields: design, negative results, and out-of-sample validation

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Abstract

Scaling processes couple strongly nonlinear kinetics to hard operational constraints, and the models available for them are structurally wrong in ways that matter. In this work the wrongness is measured rather than assumed: a classical fitted rate law mispredicts independent modern kinetic data by a factor of six at the median. We present a control architecture that treats such mismatch as a first-class design object. A hybrid soft sensor combines a physics core with a learned residual whose weight is set by its own leave-one-out credibility, so the correction fades back to the prior whenever the data has not earned it. A vertex-robust economic real-time optimizer then consumes both the correction and its uncertainty, the latter as a multiplicative constraint back-off. On a synthetic brine-line plant grounded in the primary-literature chemistry of Philippine liquid-dominated geothermal fields, the sensor reduces estimation error by 40% relative to physics alone and 48% relative to a purely data-driven model under sparse daily labels. In closed loop it is the only tested policy that remains essentially violation-free in two regimes: benign random mismatch (+33% power uplift, zero violation days) and an adverse stress set at the measured sixfold kinetic gap (0.5 post-warm-up violation days against 17.2 for the robust open loop). Two pre-registered failures, of zero-shot residual transfer and of scalar modifier adaptation against multiplicative mismatch, motivated

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the design and are reported as results. Every kinetic, solubility and sensing component is validated out-of-sample against an independent 2024 experimental dataset.
Keywords: soft sensor, hybrid modeling, real-time optimization, plant–model mismatch, geothermal energy, silica scaling, Philippines

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1. Introduction

Process models tend to be wrong precisely where constraints bind. This paper works through a case in which the wrongness can be put on a scale. The classical kinetic law for amorphous-silica polymerization, transcribed faithfully from its 1979 primary source, mispredicts the effective decay rate observed in an independent 2024 laboratory corpus by a factor of six at the median. Yet silica scaling must be managed daily and at industrial scale. In the liquid-dominated geothermal fields of the Philippines, which rank among the largest installed capacities in the world [1], dissolved silica sets the floor on reinjection temperature and therefore the ceiling on bottoming-cycle power. Mitigation practice at the Bacman- and Tiwi-type fields that motivate this study is documented and static: fixed inhibitor programs and manually managed reinjection temperatures [2–5]. Between a model too wrong to trust and a practice too static to optimize sits the control problem addressed here.

The individual ingredients are established. Machine-learned predictors of geothermal scaling exist in prediction-only form [6]. Hybrid physics–data modeling has a mature canon [7–10], soft-sensor practice under sparse labels is well codified [11], and real-time optimization under plant–model mismatch has a developed theory of modifier adaptation [12–14]. What has been missing, and what a constraint with fourth-power sensitivity punishes, is the coupling between them: an estimator that knows how much of its own learned correction to believe, and an optimizer that spends that belief as constraint margin. Two pre-registered negative results in this paper show that the coupling is not a refinement but a requirement. A residual transferred across plants actively

harms, and a scalar bias adaptation, the natural first recourse, cannot stabilize a multiplicative kinetic error on a fourth-power constraint even while it harvests spectacular nominal profit.

The contributions are fivefold. First, a credibility-weighted hybrid soft sensor whose leave-one-out shrinkage yields provable graceful degradation to its physics prior and doubles as the deployment safety interlock. Second, a vertex-robust economic real-time optimizer that consumes the sensor's correction together with its cross-validated uncertainty as a multiplicative back-off. Third, a two-regime closed-loop evaluation, spanning benign random mismatch and an adverse stress set at the measured sixfold gap, that maps a safety–harvest frontier on which the proposed loop is the only policy essentially free of violations in both regimes. Fourth, out-of-sample validation of every physics component and of the sensing pattern itself against the independent 2024 corpus, which yields a reconciliation of second-order decay with the classical fourth-order peak-rate law, a mutual solubility validation at a median of 1.05×, and a design principle we state as: identifiability precedes correction. Fifth, a fully specified and regenerable benchmark problem, built on Philippine-field chemistry, with a quartic constraint, an induction delay, a non-monotonic actuator and a ×101 temperature lever, offered to the constrained-optimization community.

Section 2 states the problem and its industrial context, Section 3 the architecture, Section 4 the in-silico results, Section 5 the out-of-sample validation, and Section 6 the limitations and what generalizes.

2. The scaling-control problem as a benchmark

2.1. Philippine industrial context

The Philippines operates among the world's largest installed geothermal capacities, dominated by liquid-dominated fields whose separated brines leave the flash plant at or near amorphous-silica saturation [1]. Polymerization of that silica sets the effective floor on reinjection temperature, and with it the ceiling on bottoming-cycle heat recovery, across fields of the Bacman and Tiwi type

whose chemistry anchors this work. The endmember brine envelope is taken from the Tiwi geochemical survey [15]; twenty years of geochemical records at Bacman document how far a single field's chemistry can wander over an operating life [16].

Mitigation practice at these fields is documented and, importantly for what follows, static. The record includes inhibitor hot-injection trials at BacMan-II [2], field tests of pH modification [3], a sustained phosphino-carboxylate inhibitor program [4], and manual management of reinjection lines [5]. These programs form the empirical baseline against which our controller is benchmarked; none of them closes a model-based loop. Economic quantities throughout are peso-denominated, with every coefficient declared as a scenario band. Where trade-press figures inform a scenario, notably the 400 kg/s Bacman-scale brine stream, they set scale only and are never fitted to or used as validation targets; the repository enforces this rule in code.

2.2. Chemistry as control structure

Read in control language, silica scaling is a constraint process with four features that together make it a demanding benchmark for economic optimization under mismatch. The classical rate correlation is a fourth-order peak-rate law in supersaturation [17], so modest errors in chemistry or temperature are raised to the fourth power exactly where the constraint binds. The rate constant is non-Arrhenius above roughly 90 °C: a low-temperature Arrhenius fit (E_a = 9.1 kJ/mol over 5–90 °C) overpredicts k at 150 °C by a factor of 3.0 (Fig. 2), which means extrapolating cold-lab kinetics to a hot brine line is quantitatively wrong rather than merely imprecise. The process carries an induction delay, with onset lagging supersaturation exposure and the induction period falling steeply in supersaturation ($P \propto s^{-4.3}$, $r^2 = 0.88$, fitted from the same primary dataset). Finally, the inhibitor actuator is non-monotonic [18]: the best organic formulation suppressed deposition by more than half near 1 ppm, and inhibition then decreased at higher dosage. Dosing is therefore a genuine optimization problem, not a saturating knob.

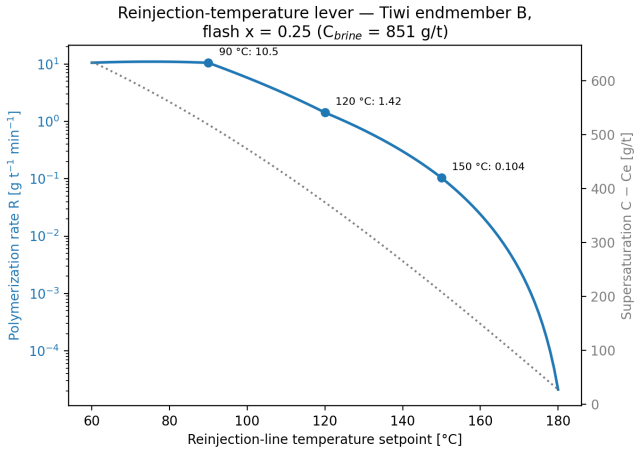


Figure 1: The reinjection-temperature lever. Uninhibited polymerization rate along the brine line versus temperature set-point for the archetype brine (Tiwi endmember B, 638 mg/kg reservoir silica [15]; flash fraction $x = 0.25$, scenario input; solubility from Gunnarsson and Arnórsson [19]; kinetics from the checksum-validated transcription of Rothbaum and Rohde [17]). Cooling from 150 to 90 °C raises the rate by a factor of 101.

Solubility sets the economic stakes of the temperature lever. Using the primary amorphous-silica correlation of Gunnarsson and Arnórsson [19] with the Tiwi archetype brine, cooling the reinjection line from 150 to 90 °C raises the uninhibited polymerization rate by a factor of about 101 (Fig. 1). Every degree of additional heat recovery is purchased at a steeply nonlinear scaling price, which is why a static policyleaves either megawatts or safety margin on the table.

3. Architecture

3.1. Synthetic-truth plant

Performance claims for an estimator or a controller mean little unless the truth they are tested against differs from the model in ways the designer did not choose to be easy. We therefore construct a synthetic plant whose physics is a structurally perturbed instance of the Section 2 core, with every deviation

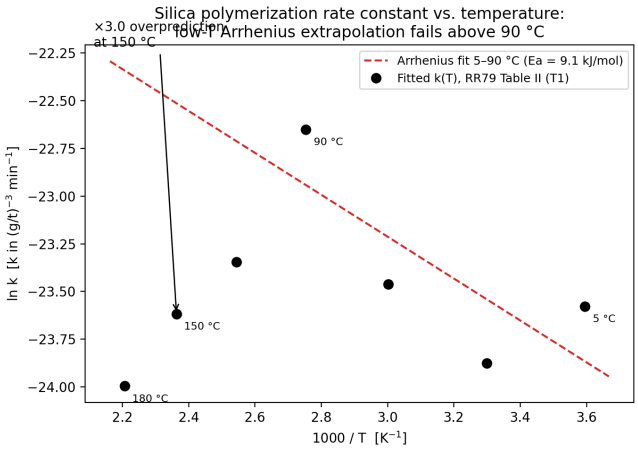


Figure 2: Fitted fourth-order rate constants $k(T)$ over 5–180 °C, regenerated from the primary transcription of Rothbaum and Rohde [17]. The dashed Arrhenius fit to the 5–90 °C nodes ($E_a = 9.1$ kJ/mol) overpredicts k at 150 °C by a factor of 3.0.

parameterized and documented. The true polymerization rate is

$$r_{\text{true}} \;=\; f_u(u)\, e^{B(T)}\, k(T)\, s^{\,n}, \qquad s = C - C_e(T)\,\beta_e, \tag{1}$$

gated by an induction clock. Here $k(T)$ and $C_e(T)$ are the literature-fitted core; $B(T)$ is a smooth log-normal bias field drawn per plant on the seven kinetic isotherm nodes ($\sigma = 0.35$); the reaction order is drawn as $n \sim \mathcal{N}(4, 0.15)$, clipped to $[3.6, 4.4]$ in keeping with the "approximately fourth order" hedge of the source data; $\beta_e \sim \mathcal{N}(1, 0.03)$ perturbs solubility; and f_u is a per-plant perturbed instance of the non-monotonic dose–response family anchored to field observations [18], with nominal suppression of about half near 1 ppm and a mild reversal on overdose. The induction period is fitted from the primary kinetic dataset itself, $\log_{10} P = a + b \log_{10} s + cT$ with $b = -4.31$ and $r^2 = 0.88$, a relation that independently reproduces the source publication's own graphical correlation. The truth maintains a supersaturation-exposure clock that the model core deliberately lacks.

Brine chemistry follows a mean-reverting Ornstein–Uhlenbeck fluctuation of endmember mixture weights around the nominal operating composition on the

Tiwi envelope [15], with a bounded flash-fraction perturbation. One modeling correction deserves open mention. An earlier revision used a pure simplex random walk, which drifts toward the mixture centroid and quietly diluted the brine from 799 to 579 mg/kg over ten days, relaxing the constraint for every controller in the study. The mean-reverting form restores stationarity (drift below 1.3% over ten days), and stationarity is now one of the repository's automated checks. The measurement layer is deliberately shift-realistic: silica grab samples every 8 h at 5% relative noise, continuous line temperature at $\sigma = 0.3\,^{\circ}\mathrm{C}$, and a single noisy daily observation of the trailing-24-h mean rate (10% relative noise, standing in for a coupon or turbidity proxy). A fourteen-day episode thus provides 42 concentration samples and 14 rate labels against 2,016 hidden truth values. This is the sparse-supervision regime the architecture is built for. A step-wise simulation interface reproduces the batch simulator's truth path exactly, verified to machine precision, so that closed-loop experiments can let the controller decide each day from what it has actually observed.

3.2. Credibility-weighted hybrid soft sensor

The sensed quantity is the trailing-24-h mean polymerization rate: the quantity that accumulates scale, and the one the optimization layer constrains. The sensor is a parallel hybrid [8, 9],

$$\hat{r} \;=\; \mathrm{expm1}\big(\log(1{+}r_{\mathrm{core}}) \;+\; \lambda\, g_\theta(x)\big)_{+}, \tag{2}$$

in which r_{core} is the trailing mean of the kinetic core combined with the nominal literature dose–response, and g_θ is a gradient-boosted residual [20] in $\log(1{+}\cdot)$ space over causal trailing-window aggregates of observables only. A code-level guard raises an error if any truth-side variable reaches the feature set. Training is per plant, on that plant's own sparse daily labels, under strict temporal splits, which is the standard soft-sensor protocol [11].

The protocol is itself a result. Our first evaluation asked the residual to generalize zero-shot to unseen truth realizations, and the hybrid lost to both a pure-physics and a pure-data baseline. The diagnosis is structural rather than

statistical. Per-plant mismatch draws are unpredictable before any data from that plant exists, so a zero-shot split tests a claim no soft sensor makes; a soft sensor is trained on its plant's history and predicts forward in time, where the plant's own bias is identifiable. We report this negative result because it motivates the two protective mechanisms that follow. The residual weight λ in Eq. (2) is set by leave-one-out (LOO) credibility on the training labels,

$$\lambda^{*} = \mathrm{clip}_{[0,1]} \frac{\langle e, \tilde{r} \rangle}{\langle \tilde{r}, \tilde{r} \rangle}, \qquad \lambda = 0 \text{ unless the LOO SSE improves by at least } 15\%, \tag{3}$$

where e collects the residual targets and $\tilde{r}$ their LOO predictions. The correction is used only to the extent that it demonstrates out-of-sample signal, and the hybrid falls back gracefully to its physics prior as labels become few or uninformative. The same LOO computation yields the residual's prediction-error standard deviation σ_{LOO}, which is exported to the optimization layer as the sensor's own statement of uncertainty.

3.3. Uncertainty-consuming robust economic RTO

Following the two-layer rationale of steady-state real-time optimization [14], the decision variables are the reinjection temperature set-point and the inhibitor dose, (T_r, u), refreshed daily. The objective is heat-recovery revenue minus inhibitor cost, with every economic coefficient declared as a scenario parameter (peso-denominated; Section 2.1). The scaling constraint is enforced on every vertex of the brine-chemistry envelope at the adverse flash bound,

$$\max_{T_r,\, u} \; J(T_r, u) \quad \text{s.t.} \quad \hat{r}_v(T_r, u)\, e^{\kappa \sigma_{\mathrm{LOO}}} + \varepsilon \;\le\; r_{\max} \quad \forall v \in \mathcal{V}, \tag{4}$$

and the problem is solved by dense grid search. The choice is deliberate: on a bounded two-dimensional decision space the grid is global, deterministic and auditable by any reader, and the vectorized implementation is consistency-checked against the reference solver at start-up. Feedback enters in two forms that the results deliberately contrast. A scalar modifier-adaptation bias ε, updated from daily innovations [12, 13], is retained as the natural feedback baseline. In the proposed coupling, the trained sensor of Section 3.2 becomes the constraint

Table 1: Estimation ablation on unseen test windows: RMSE in log(1+·) rate space (mean ± sd over 8 plants) and the fraction of estimates within a factor of two of the hidden truth. Source: `results/table1_summary.csv`.

Arm	$\mathrm{RMSE}_{\log 1p}$	vs. physics	within ×2
Physics core	0.196 ± 0.110	—	41%
Pure ML	0.227 ± 0.086	+16%	37%
Hybrid (this work)	$\mathbf{0.117 \pm 0.045}$	**−40%**	**63%**

model $\hat{r}_v$ itself, queried at steady state across the decision grid, its correction shrunk by λ and backed off by its own $\kappa\sigma_{\mathrm{LOO}}$ with $\kappa = 1$. During warm-up, before the sensor has earned credibility, feedback may only tighten the constraint ($\varepsilon \geq 0$). The justification for that rule is empirical and appears in Section 4.3.

4. In-silico results

4.1. Estimation under sparse labels

Eight independent truth realizations were simulated for 90 days each. Sensors were trained on the first 60 days of labels and evaluated on the final 30, following the per-plant temporal protocol. Table 1 summarizes the ablation triple. The hybrid reduces error by 40% against its own physics prior and by 48% against the pure-data baseline. The label-scarcity sweep in Fig. 3 sharpens the point. With ten days of labels the hybrid (0.161) already outperforms the physics core given any amount of data (0.196), while the pure-data arm needs the full sixty days to approach what the hybrid achieves in ten. The transfer experiment then bounds the claim honestly: a residual carried to a foreign plant along with its donor credibility does active harm (0.317 against the physics prior's 0.155). This is exactly why λ must be earned from the target plant's own labels. At zero labels $\lambda = 0$ and the hybrid simply is its prior. The credibility mechanism is the deployment safety interlock, not decoration.

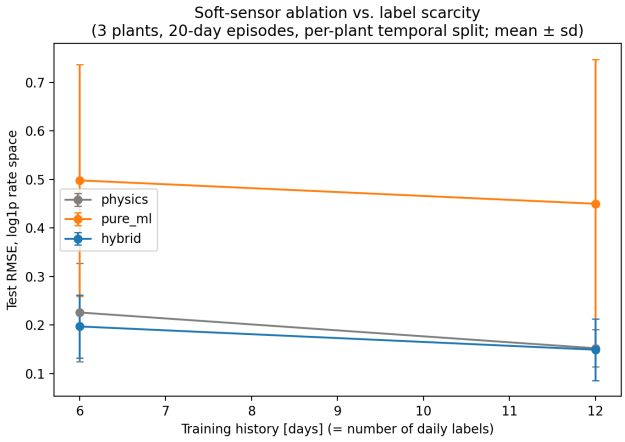


Figure 3: Test RMSE versus training-history length, equal to the number of daily labels; mean ± sd over 8 plants. Source: `results/scarcity_sweep.csv`.

Table 2: Closed-loop benchmark in the benign regime (random per-plant mismatch; 6 plants × 45 d). Violation days count days whose realized trailing-24-h mean rate exceeded $r_{\max}$. Source: `results/table2_summary.csv`.

Arm	Gross MW	Uplift	Viol. days (max)	Inhib. PHP/MWh
Baseline (fixed)	5.03	—	0.00 (0)	85.9
RTO, no feedback	7.12	+41.7%	0.17 (1)	51.6
RTO + scalar MA	9.26	+84.1%	5.33 (12)	41.2
RTO + hybrid, κ=1	6.71	+33.4%	**0.00 (0)**	53.5

4.2. Closed loop under benign mismatch

Table 2 reports the four-arm closed-loop benchmark on six plants over 45 days each: a fixed empirical-practice baseline at 150 °C and 1 ppm continuous dosing, the pattern of documented field programs [4]; the vertex-robust RTO without feedback; the scalar-MA loop; and the proposed hybrid-in-the-loop controller. Three observations follow. Even the robust open-loop RTO leaks: one plant, one violation day, because an envelope over brine chemistry does not certify against kinetic draws. The scalar-MA loop harvests spectacularly and unsafely, posting +84% uplift at an average of 5.33 violation days per plant

Table 3: Adverse stress at the measured sixfold kinetic gap (single-factor; 6 plants × 45 d). "Post-w." counts violation days after the 8-day warm-up. Source: `results/table2b_summary.csv`.

Arm	Uplift	Viol. days	Post-w.	Net kPHP/h
Baseline (fixed)	—	0.00	0.00	27.2
RTO, no feedback	+41.7%	20.67	17.17	38.8
RTO + scalar MA	+37.0%	16.50	14.17	37.5
RTO + hybrid, κ=0	+21.0%	4.33	2.17	33.1
RTO + hybrid, κ=1	+6.2%	**2.67**	**0.50**	29.1

and twelve on the worst plant, in the benign regime. The hybrid controller is violation-free on all six plants; its +33% uplift sits 8 percentage points below the no-feedback RTO, and that gap is the premium of the $\kappa\sigma$ back-off. It is the price of insurance, quantified.

4.3. Adverse-kinetics stress

The stress magnitude is a measurement, not a choice. The out-of-sample validation of Section 5 finds a sixfold median gap between the fitted core and independent modern kinetics, so the stress sets the plant's true rate six times above the model as a deterministic single-factor perturbation with all random draws zeroed. Table 3 reports the pre-registered outcome. The no-feedback RTO violates chronically, on 17.2 of 37 post-warm-up days: its vertex envelope covers brine uncertainty, not kinetic mismatch. The scalar-MA loop barely improves on open loop (14.2), and the mechanism is now precise. An additive bias correction learned at one operating point cannot stabilize a multiplicative error on a quartic constraint; the absolute error it measures is small when the plant runs hot and explodes when the optimizer moves cold, so the loop hunts and violates in a limit cycle. The hybrid constraint, being state-dependent, extrapolates structurally. Even without margin (κ=0) it cuts post-warm-up violations eightfold, to 2.17 days. Our pre-registered target for that arm was zero, and it was not met; we report the miss. The uncertainty back-off (κ=1)

closes the remainder to 0.50 days at a further harvest cost, which we read as an honest frontier point: under a sixfold kinetic error there is little harvest that is safe to take. We also observe the optimizer substituting the dose lever for the temperature lever once the latter becomes risk-expensive, with inhibitor intensity falling to 38–45 PHP/MWh on several stressed plants. Across both regimes the conclusion is a two-by-two. Open-loop robustness is safe but blind, scalar adaptation is reactive but unsafe, and only the credibility-weighted hybrid with back-off remains essentially violation-free in both.

4.4. Global sensitivity and economic ranges

A Sobol study ($N = 512$, seeded, 9,216 evaluations) spans the declared uncertainty set: a kinetic band equal to the measured sixfold gap, a solubility band equal to the measured endpoint spread $[0.90, 1.37]$, and declared scenario bands for flash, dose–response, $r_{\max}$ and economics (Fig. 4). Kinetic model-family uncertainty dominates the economics, with total-order indices of 0.41, 0.57 and 0.33 on gross power, optimal temperature and net objective. The quantity the sensing layer reduces is the quantity that moves the money. The optimal dose is governed instead by dose–response knowledge ($S_T = 0.67$), which identifies inhibitor characterization as the second identification target. Feasibility holds at 100% across the set, so the declared uncertainty never strands the optimizer, and the gross-power optimum spans 6.0/11.7/17.6 MW ($q_{05}/q_{50}/q_{95}$). Finally, re-valuing the committed closed-loop trajectories over the economics envelope attaches ranges to every peso figure. The enabling assumption is verified rather than assumed: the optimum moves zero grid steps across all eight envelope corners, with inhibitor cost below 15% of revenue everywhere, so uplift percentages and violation counts are untouched. The safety conclusions of Tables 2 and 3 do not depend on economic assumptions.

5. Out-of-sample validation on independent data

Every physics component of the architecture, and the sensing pattern itself, is confronted with an experimental dataset it was never fitted to: the supple-

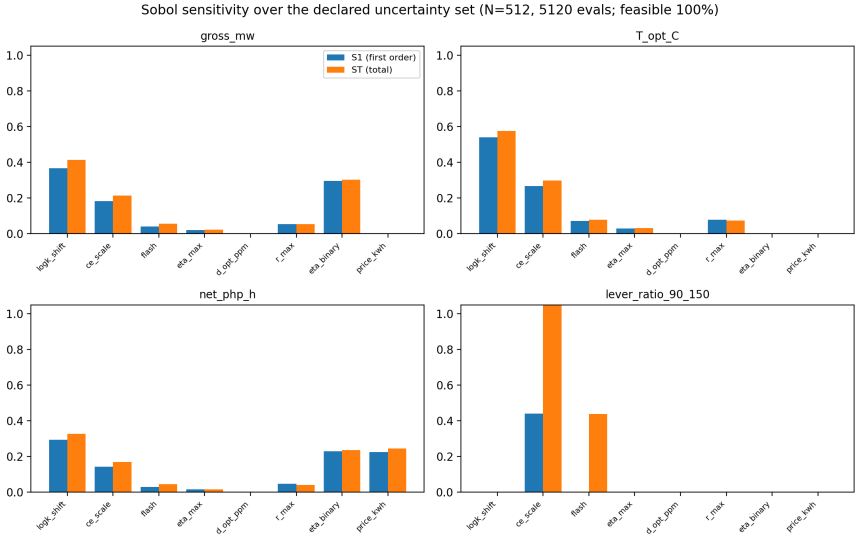


Figure 4: First- and total-order Sobol indices over the declared uncertainty set. Source: `results/sensitivity_sobol.csv`.

mentary data of Scott et al. [21], read with the parameterization corrections of Scott et al. [22]. The corpus comprises 61 monomeric-silica decay experiments and 1,275 points spanning 23–80 °C, pH 2.1–10.0 and ionic strength 0.01–0.53 mol/kg. Because the source is machine-readable, manual transcription protocols give way to automated parser integrity checks: unit-header pinning, monotonicity, separation of replicate runs listed under duplicated identifiers, and flagging of one colloid-regime outlier that falls outside the dissolved-monomer scope.

5.1. Kinetics: what a 1979 law does and does not predict

Two properties of the fitted core are tested separately, because they fail differently. Consider first the functional form. Linearizing the integrated n-th-order decay law and sweeping $n \in \{2, \ldots, 6\}$ per experiment shows that single-curve monomer decay in the modern data is cleanly second order in supersaturation: median $R^2 = 0.992$ at $n = 2$, degrading monotonically as the assumed order rises, with best-fit order 2 in all eleven qualifying near-neutral ex-

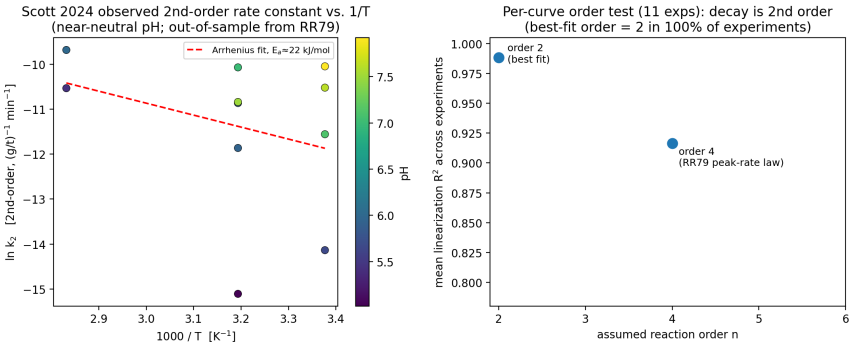


Figure 5: Out-of-sample kinetic validation against Scott et al. [21]. Left: observed second-order rate constants versus inverse temperature for the near-neutral experiments, with the Arrhenius trend. Right: mean linearization R^2 versus assumed reaction order; single-curve decay is second order, while the fourth-order form is a peak-rate law. Source: `results/kinetics_validation.csv`.

periments (Fig. 5). This does not contradict the source of the core. The classical "fourth order" is a correlation of maximum rate against initial supersaturation across separate batches, a peak-rate law, whereas the integrated decay of a single curve follows Rimstidt–Barnes-type second-order kinetics [23]. The two describe different observables, and we report both rather than forcing either onto data that says otherwise. Now the rate level. The observed second-order constants follow an Arrhenius trend with $E_a \approx 22$ kJ/mol, and the effective decay constant implied by the fitted core differs from observation by a factor of about six at the median. That number does double duty in this paper. It calibrates the adverse stress of Section 4.3, and it is the measured size of the gap the learned residual exists to absorb: no single transcribed rate law reproduces independent modern data.

5.2. Solubility: mutual validation across 24 years

The plateau of a near-complete decay experiment is an independent measurement of amorphous-silica solubility. Sixteen near-neutral experiments that consumed at least 70% of their initial supersaturation plateau at 0.90–1.37 times the Gunnarsson and Arnórsson [19] correlation, with a median of 1.05. The 2000

correlation and the 2024 endpoints therefore validate each other in both directions, and the measured band $[0.90, 1.37]$ becomes the solubility axis of the sensitivity study in Section 4.4.

5.3. The sensing pattern on real curves: identifiability precedes correction

Finally, the field sensing pattern is replayed on laboratory truth. Each qualifying experiment is treated as a plant, interval-average observed rates as the hidden truth, the fitted core as the prior, and only the first K intervals as labels. Three arms with honestly counted parameters are compared: the core alone; a one-parameter scalar hybrid, a fitted log-level offset that is deployable from a single label, where a data-only fit cannot operate at all; and a two-parameter affine hybrid whose slope in $\ln s$ is, by construction, an order correction. Second-order truth against a fourth-order core predicts $b \approx -2$.

The raw affine arm is retained as a cautionary ablation. Early labels span almost no state (a label $\ln s$ spread of 0.047 against a test span of 2.61), so unregularized slopes explode, with b ranging from -230 to $+769$, and the arm performs far worse than no correction at all. Applying the same credibility mechanism as the field sensor, a leave-one-out selection between the scalar and a clipped affine, the correction collapses to the safe scalar whenever the labels have not earned the slope ($K \leq 3$). At $K = 5$ it engages in 9 of 21 experiments with median $b = -2.50$, independently re-measuring the order-2 finding, and wins outright with a median RMSE of 2.59 against 3.04 for the core (Fig. 6). The principle this crystallizes, identifiability precedes correction, also explains the field results. A plant regulated near an operating point presents a level-dominated, label-covered correction problem, which is exactly where hybrid residuals win; wide-range extrapolation demands state-spanning labels, and a credible architecture must know which regime it is in.

6. Discussion and limitations

Evaluation posture.. The closed-loop results are in silico. The plant is synthetic, though every physical ingredient, from solubility and kinetics to the

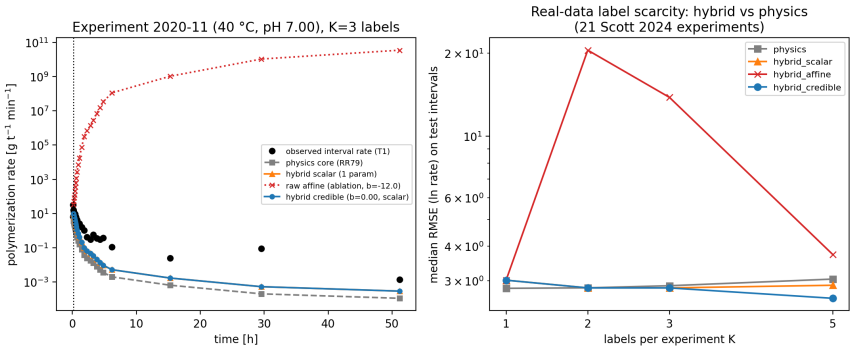


Figure 6: The sensing pattern on real decay curves. Left: a representative experiment, showing the observed interval rates, the diverging physics core, the raw-affine explosion, and the credible hybrid. Right: median test RMSE versus labels per experiment K for all arms, on a log scale. Source: `results/sensor_validation_scott.csv`.

brine envelope and the actuator family, is grounded in primary literature and, wherever independent data exists, validated out-of-sample component by component (Section 5). We regard component-wise real-data validation combined with a parameterized-mismatch closed loop as the honest intermediate between simulation-only studies and a field trial. A field pilot is the natural next step, and it would require exactly the sparse instrumentation this architecture is designed around.

Constraint units.. The scaling constraint is expressed as a polymerization (monomer-depletion) rate rather than a wall-deposition thickness rate. The molecular-deposition treatment of Weres et al. [24], transcribed under the same integrity protocol (its printed checkpoints reproduce, including the 3.0 kcal/mol effective activation energy; one internal inconsistency in its Eq. [4c] is documented rather than silently repaired), closes the unit gap. Along the envelope where the constraint is active, $r_{\max} = 0.5\ \mathrm{g\,t^{-1}\,min^{-1}}$ corresponds to a Weres-implied molecular-deposition wall growth, a lower bound on total scale growth, of about 0.3 mm/yr at the optimized operating point and 0.04 to 1.6 mm/yr across 90–175 °C; the four-fold $r_{\max}$ band of the sensitivity study spans that calibration.

Scenario fencing.. Trade-press figures inform scenario scale only. The rule is enforced mechanically in the repository, and every peso and megawatt headline carries an envelope range with decision invariance verified rather than assumed.

Design constants.. The back-off gain is fixed at $\kappa = 1$. Its ablation over $\kappa \in \{0, 1\}$ brackets the safety–harvest trade under stress, and σ_{LOO} is already data-adaptive. A continuous κ sweep is the tuning study we would prioritize next.

Scope boundaries.. High-pH brines require silica ionization corrections that are deferred here; the speciation layer exists in the repository but is not exercised in the loop. The identifiability principle of Section 5.3 is likewise a scope statement rather than a weakness: corrections are used only where labels can support them, and the architecture knows which regime it is in.

What generalizes.. Nothing in Eqs. (2)–(4) is specific to silica. The pattern, a physics prior corrected only to its earned credibility and feeding a constraint backed off by the correction's own cross-validated uncertainty, applies to any hard-constrained process operated against a structurally wrong model: fouling, crystallization, corrosion, membrane scaling. Geothermal silica supplies an unusually clean and unusually demanding instance, with quartic constraint sensitivity, an induction delay, non-monotonic actuation, and a measured sixfold model-family gap, and that is why we offer it as a benchmark.

7. Conclusions

We posed geothermal silica scaling as what it is for a control engineer: a hard-constrained economic optimization run against a model known to be structurally wrong, with the size of the wrongness measured at a sixfold median gap against independent modern kinetics rather than assumed. The architecture answers mismatch with two coupled mechanisms, a hybrid soft sensor whose learned correction is weighted by its own leave-one-out credibility, and a vertex-robust economic real-time optimizer that consumes both the correction and its uncertainty as a constraint back-off. Under sparse daily labels the sensor cuts

estimation error by 40% against its physics prior and by 48% against a pure-data baseline. In closed loop it is the only tested policy that remains essentially free of constraint violations in both regimes: benign mismatch, at +33% power uplift with zero violation days on six of six plants, and a stress set at the measured sixfold gap, at 0.50 post-warm-up violation days against 17.2 for the robust open loop. Two pre-registered failures shaped the design and are reported as results. Zero-shot residual transfer harms, which makes earned credibility the deployment interlock; and scalar modifier adaptation cannot stabilize a multiplicative error on a quartic constraint, which makes state-dependent correction necessary rather than merely preferable. Every physics component, and the sensing pattern itself, was validated out-of-sample. That exercise returned two findings of independent geochemical interest, that single-curve decay is second order against the classical fourth-order peak-rate law and that the 2000 solubility correlation and the 2024 endpoints mutually validate at a median of 1.05×, together with one transferable design principle: identifiability precedes correction.

Data and code availability

The complete repository, comprising the primary-data transcriptions with integrity checks, all models, experiments and figures, regenerates every result through a single gate script (94 automated checks across 13 modules, deterministic seeds throughout). An archival release is planned upon acceptance. A third-party author-provided reference script is excluded pending its author's permission.

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checking, which is not redistributed with this work. This work received no external funding.

Appendix A. Data transcription and integrity protocol

Primary numerical sources enter the repository only through a transcription protocol: tables are transcribed to CSV against the publisher's text layer, validated against printed derived quantities where the source provides them (for the kinetic dataset of Rothbaum and Rohde [17], the printed $R \cdot P^{1.5}$ checksum column validates 33 of 33 retained rows; one flagged row is excluded), and tagged with an evidence tier: T1 for peer-reviewed primary sources, T2 for conference or institutional material, and T3 for trade press, which may motivate scenario scale only and is never fitted, a rule enforced in code. Machine-readable sources replace manual transcription with automated parser integrity checks (Section 5).

Appendix B. Reproducibility

A single gate script executes the self-checks of all twelve modules (94 checks at the time of writing) and refuses a green status unless every check passes; the repository convention is that no commit is made against a red gate, and all figures and result tables regenerate from source and raw transcriptions. All stochastic components are seeded, including the Sobol sampler, whose default scrambled quasi-random sequence is pinned for run-to-run reproducibility. The Python environment is pinned in `requirements.txt`; principal dependencies are NumPy/SciPy/pandas, LightGBM [20], SALib [25], and PHREEQC via phreeqpython [26] for the (deferred) speciation layer.

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